Example 1 Creating threads by extending Thread class

```
package com.cait.example1;

class MyThread extends Thread {

    @Override
    public void run() {
     //ticket booking
        for (int i = 1; i <= 5; i++) {
            System.out.println(
                Thread.currentThread().getName() + " data
- " + i
            );
        }
    }
}

public class ThreadMain {

    public static void main(String[] args) {

        MyThread t1 = new MyThread(); // thread1
        MyThread t2 = new MyThread(); // thread2
        MyThread t3 = new MyThread(); // thread3

        t1.setName("Thread-1");
        t2.setName("Thread-2");
        t3.setName("Thread-3");


        t1.start();
        t2.start();
        t3.start();

       /*
        t1.run();
        t2.run();
        t3.run();
        */
    }
}
```

Example 2 – Creating thread by implementing the Runnable interface

```
package com.cait.example2;


class Railway implements Runnable {
    private String userName;

    public Railway(String userName) {
        this.userName = userName;
    }

    @Override
    public void run() {
        System.out.println("Ticket booking started for: "
+ userName);
        System.out.println("Ticket booked successfully
for: " + userName);
    }
}

public class A_TicketBooking {
    public static void main(String[] args) {
        // Create Runnable objects
        Railway user1 = new Railway("Alice");// not the
thread object
        Railway user2 = new Railway("Bob");// not the
thread object

        // Wrap them inside Thread objects
        Thread t1 = new Thread(user1);//thread 1
        Thread t2 = new Thread(user2);//thread 2

        // Start threads
        t1.start();
        t2.start();
    }
}
```

Example 3 – sleep method

```
public class A_AlarmSimulation
```

```
{
    public static void main(String[] args)
    {
        System.out.println("Alarm set for 5 seconds from
now...");
        //1000 ms - 1 sec
        try {
            Thread.sleep(5000); // wait for 5 seconds
        }
        catch (InterruptedException e) {
            System.out.println("Alarm was interrupted!");
        }
        System.out.println("Ring Ring! Wake up!");
    }
}
```

Example 4 – sleep method

```
class AlarmThread extends Thread {

    @Override
    public void run() {

        System.out.println("Alarm started...");
      //t1 will sleep for 5 seconds and thread1 resumes
after 5 seconds
        try {
            Thread.sleep(5000); // AlarmThread sleeps for
5 seconds
        } catch (InterruptedException e) {
            System.out.println("Alarm was interrupted!");
        }

        System.out.println("Ring Ring! Wake up!");
    }
}

public class B_AlarmSimulation {

    public static void main(String[] args) {

        AlarmThread t1 = new AlarmThread();//thread object

        t1.start();
```

```
        System.out.println("Main thread is doing some
other work...");
    }
}
```

Example 5 – join method

```
//verify join process is guranteed one or not.


class CookingTask extends Thread {
    private String taskName;

    public CookingTask(String taskName) {
        this.taskName = taskName;
    }

    @Override
    public void run() {
        System.out.println(taskName + " started.");

        // Instead of a loop, just print directly
        System.out.println(taskName + " in progress...
Step 1");
        System.out.println(taskName + " in progress...
Step 2");

        System.out.println(taskName + " completed.");
    }
}

public class A_JoinMethod {
    public static void main(String[] args) {
     //thread1 is Cutting Vegetables
        CookingTask cuttingThread = new
CookingTask("Cutting Vegetables");//thread1 object
        //thread2  is Boiling Vegetables
        CookingTask boilingThread = new
CookingTask("Boiling Vegetables");//thread2 object
```

```
        cuttingThread.start();

     // main thread waits until cuttingThread completes
its process
        try {
         cuttingThread.join();
        } catch (InterruptedException e) {
            e.printStackTrace();
        }

        boilingThread.start(); // start boiling only after
cutting
    }
}
```

Example 6 – setName method

```
class MyThread extends Thread {

    @Override
    public void run() {

        // Get the currently executing thread
        Thread currentThread = Thread.currentThread();

        // Get the name of the current thread
        System.out.println("Currently executing thread: "
                + currentThread.getName());
    }
}

public class ThreadNameMain {

    public static void main(String[] args) {

        MyThread t1 = new MyThread();
        t1.setName("Thread 1 ");
        t1.start();


        MyThread t2 = new MyThread();
```

```
        t2.setName("Thread 2");
        t2.start();

    }
}
```